

ROBOFEST- GUJARAT 6.0

Guidelines for submission of Ideation Proposal
Senior category

All teams must be present and ready to perform their challenge. The committee may assign any team to the task at random. Any kind of absence and lack of readiness will be subject to suitable actions, including the deduction of score/disqualification.

Students are required to perform their task and answer the questions. Mentors should not be involved in the task. The committee may allow mentors during question-answer if it feels necessary, depending on the situation.

Aerial Robotics: Minefield Navigation Challenge

Game Description:

In this senior-level challenge, teams will design and deploy a swarm of autonomous aerial vehicles to help a “person-at-risk” safely cross a 15 m x 60-meter-long minefield within 10 minutes. The teams must deploy a minimum of 3 miniature drones to qualify in the competition. Ideally, teams can use 3 to 4 miniature drones to have an effective swarm formation. However, teams can use more than 4 drones also. There won't be any restriction from GUJCOST regarding the upper limit of the number of drones to be deployed. The mission demands advanced aerial coordination, obstacle avoidance, swarm communication, and safe path mapping without external control systems. This challenge promotes real-world relevance in defense, disaster response, and humanitarian assistance.

Challenge Objective:

- Safely navigate a person-at-risk from pre-defined Start Zone to Exit Zone across a 15 m x 60-meter “minefield”.
- Each team must launch the autonomous miniature aerial vehicles (under 750 gram each).
- Navigation: The use of Global Navigation Satellite System (GNSS)/ Global Positioning System (GPS) or any external positioning system is strictly prohibited. All localization, navigation, mapping, and decision-making shall be performed autonomously using onboard sensors and onboard computation only.
- Inter-Drone Communication: Communication between drones of the same team is permitted solely for swarm coordination (e.g., task allocation, formation control, and status sharing) using license-exempt wireless bands in accordance with the prevailing WPC, Government of India regulations.
- External Data Links: No communication with external computers, cloud servers, base stations, or remote operators shall be permitted during the competition. Remote piloting, external navigation, off-board computation, or mission updates are strictly prohibited.
- Emergency Kill Switch: An independent emergency kill switch is mandatory for all teams. The kill-switch link shall be used exclusively for flight termination or motor disarming and shall not carry any navigation, control, mission, or payload-related commands.
- Compliance: The Technical Advisory Committee (TAC) may inspect the communication architecture and software of the drones during technical scrutiny. The decision of the TAC regarding compliance shall be final and binding.
- The drones must:
 - Map the underground mines
 - Avoid redundancy in mine detection.
 - Generate and communicate a 1-meter clearance safe path.

- Mark the safe path visually (LED, coloured powder, or markers).
- Track the human and re-route if the path changes.

The maximum permissible All-Up Weight (AUW) of the drone, excluding the mission payload, shall be 750 g (± 10 g tolerance). The AUW shall include the airframe, battery, propellers, motors, ESCs, flight controller, onboard computer/compute board, sensors, camera(s), communication modules, wiring, landing gear, and payload carrying mechanism.

All measurements shall be carried out during technical scrutineering using the official weighing balance, and the decision of the Technical Advisory Committee (TAC) shall be final.

Challenge Tasks:

1. Take-off and Human Gesture Recognition
 - Drones activate on hand/voice command of the “person-at-risk”.
 - Must recognize simple gestures (forward, pause, scan left/right).
2. Mine Detection and Mapping
 - Use onboard optical/IR sensors to identify mine locations.
 - Create a digital map stored in local memory for coordination.
3. Swarm Coordination
 - Coordinate with other drones to avoid rechecking the same area.
 - Each drone scouts a separate section of the field.
4. Safe Path Marking
 - Highlight safe path using visual cues (LEDs or coloured chalk/powder).
 - Maintain a 1-meter safety buffer from detected mines.
5. Path Communication and Human Tracking
 - Communicate the path to “person-at-risk” visually (arrow pattern or corridor lights).
 - Dynamically update the path if the human deviates or the minefield changes.
6. Mission Completion
 - Person-at-risk (with helmet and eye protection) crosses safely under drone guidance within 10 minutes.
 - The drone swarm must land autonomously once the mission ends.

Game Field Design:

- Field Size: 15m width $\times$ 60m length.
- Start Zone (1m $\times$ 15m): Drone deployment and human starting area.
- Minefield Zone (58m $\times$ 15m): Contains approx. 40 randomly placed “mines” (plastic disks or markers).
- Exit Zone (1m $\times$ 15m): Finish line for the person-at-risk.
- Marked Zones:
 - Mines simulated using flat discs (20–30 cm radius).
 - Nearly 25% of the ‘mines’ (flat discs) will be buried slightly inside the earth (sand or mud or soil within ~ 30 -50 mm depth, i.e. beneath the soil surface. However, those won’t be visible on the surface.
 - To ensure fair and achievable detection using onboard optical or vision-based sensors, each buried mine shall be indicated by a standardized surface marker (e.g., a coloured peg, painted ring, disturbed soil pattern, or predefined visual cue) placed directly above or adjacent to the buried mine. Teams shall detect these surface cues using onboard computer vision algorithms only. The use of external sensors or manual intervention shall not be permitted.
 - Rest $\sim 75\%$ of the ‘mines’ will be on-ground (terrestrial), having different non-reflecting opaque colours.
 - Each mine shall have a 1.0 m horizontal exclusion zone measured from the centre of the mine marker.
 - Obstacles like poles (5 m height) and trees may be present. The exact number and locations of the obstacles shall remain undisclosed until the commencement of the competition run to evaluate the autonomous navigation capability of the swarm.

Regarding SHAPE, SIZE & LOCATION of the mines: The size of the mines can vary from 20 cm to 30 cm diameter. But mines may not be of same size. Shape will be mostly circular with a few can be elliptical as well. The exact number of circular and/or elliptical mines will not be disclosed.

Mines will be of plastic material of different colour.

Stage 1: Proof of Concept (PoC) – Guidelines

Objective: To evaluate the core technical capability of teams in:

- Autonomous navigation
- Sensor integration
- Basic AI/decision-making
- Drone stability and control

Game Field Design:

- Field Size: 10m width × 40m length.
- Start Zone (1m × 10m): Drone deployment and human starting area.
- Minefield Zone (38m × 10m): Contains approx. 20 randomly placed “mines” (plastic disks or markers).
- Exit Zone (1m × 10m): Finish line for the person-at-risk.

Challenge Description (PoC Stage)

Teams must demonstrate a **scaled-down version** of the final problem with simplified conditions.

Tasks:

1. **Autonomous Take-off and Stabilization**
 - Drone must take-off autonomously and hover stably.
2. **Obstacle Detection & Avoidance**
 - Navigate through a predefined path with 3–5 static obstacles like poles or trees.
3. **Basic Target Detection**
 - Identify and mark at least 5 “simulated mines” (visible markers).
4. **Path Indication**
 - Generate a simple safe path using LEDs or visual indicators.
5. **Basic Human Interaction**
 - Execute at least one command (gesture/voice/button-based trigger).

Technical Constraints

- Max drone weight: 500 grams (excluding compute payload)
- Only onboard processing allowed (no external control except emergency override)
- Minimum 1 drone (swarm optional at PoC stage)

Evaluation Criteria (PoC)

Criteria	Marks
Stability & Control	15
Obstacle Avoidance	20
Detection Accuracy	20
Basic AI/Logic	15
Path Visualization	10
Innovation	10
Presentation	10

Stage 2: Grand Finale – Guidelines

Objective: To demonstrate a **fully integrated autonomous aerial system** solving a complex real-world problem using:

- Swarm robotics
- AI-based mapping
- Dynamic decision-making

Challenge Description (Final Stage)

Autonomous Minefield Navigation & Human Guidance

Teams must deploy a **swarm of minimum-3 drones**.

- Detect hidden mines (not visible)
- Map the terrain autonomously
- Create and communicate a safe path
- Guide a “person-at-risk” safely across the field

Game Field Design:

- Field Size: 15m width × 60m length.
- Start Zone (1m × 15m): Drone deployment and human starting area.
- Minefield Zone (58m × 15m): Contains approx. 40 randomly placed “mines” (plastic disks or markers).
- Exit Zone (1m × 15m): Finish line for the person-at-risk.

Advanced Tasks

1. **Swarm Deployment**
 - Coordinated multi-drone operation at a height of 20 ft or above
 - Formation of any geometric shape and retain it for 3 minutes (minimum) (3 min duration is addition to the mission completion time)
2. **Autonomous Mapping**
 - Generate real-time minefield map
3. **Dynamic Path Planning**
 - Ensure 1-meter safe clearance
 - Re-route in case of deviation
4. **Human Tracking & Guidance**
 - Guide user using visual cues
5. **Mission Completion**
 - Safe crossing within defined time
 - Autonomous landing

Field Design (Enhanced)

- Field Size: ~15m × 60m
- Minefield Zone: Random hidden objects
- Obstacles: Poles/walls and trees
- Environmental disturbances like wind may be present

Technical Requirements

- Fully autonomous system
- Swarm coordination mandatory
- No external navigation (GPS-denied environment optional)
- Real-time decision making required

Evaluation Criteria (Grand Finale)

Task	Points
Safe takeoff and swarm activation	10
Gesture/voice command recognition	10
Swarm formation (in the sky)	10
Mine detection & correct mapping	25
Safe path creation & marking	15
Person crosses safely	20
Time Bonus	10
Collision or unsafe proximity	-10
Drone crash or surface contact	-20

TOTAL SCORE **100**

Only 1 re-run permitted with due deduction of penalty.

Tie Breaker

1. Maximum challenges completed.
2. Minimum penalty points.
3. Fastest completion time.
4. Decision of the Technical Advisory Committee (TAC) shall be final.

Note: The guidelines related to the eligibility, timelines, formats of submissions, prizes and conduct are published on the Robofest Gujarat Webportal and shall be binding to all participating teams.